Quiz 2, 2021/2022 ME 265 THERMODYNAMICS I

INSTRUCTIONS

Select the correct ANSWER and check your answer appropriately. Questions 1 -43 will attract one mark for each question answered correctly. Questions 44 – 62 will attract 3 marks for each question answered correctly.

CYCLIC HEAT ENGINE

1. The first law of thermodynamics doesn't tell us whether a thermodynamic process is feasible or not.
a) True
b) false

2. In a cyclic heat engine there is
a) net heat transfer to the system and net-work transfer from the system
b) net heat transfer from the system and net-work transfer to the system
c) depends on the conditions of cycle
d) none of the mentioned

3. A thermal energy reservoir (TER) is a large body of
a) small heat capacity
b) large heat capacity
c) infinite heat capacity
d) none of the mentioned

4. A TER which transfers heat to system is called ____ and one which receives heat is called ____
a) source, sink
b) sink, source
c) sink, sink
d) source, source

5. Which if the following statements are true for a mechanical energy reservoir (MER)?
a) it is a large body enclosed by an adiabatic impermeable wall
b) stores work as KE or PE
c) all processes within an MER are quasi-static
d) all of the mentioned

KELVIN – PLANCK STATEMENT

6. If heat rejected from the system Q_L is zero, then
a) net-work=Q_H and efficiency=1.00
b) heat is exchanged only with one reservoir
c) it violates the Kelvin-Planck statement
d) all of the mentioned
7. A heat engine has to exchange heat with ___ energy reservoir at ___ different temperatures to produce net-work in a complete cycle.
a) One, one
b) one, two
c) two, two
d) none of the mentioned

8. The second law is not a deduction of the first law.
a) True
b) false

9. According to Clausius statement
a) it is impossible to construct a device that can transfer heat from a cooler body to a hotter body without any effect
b) it is impossible to construct a device that can transfer heat from a hotter body to a cooler body without any effect
c) it is possible to construct a device that can transfer heat from a cooler body to a hotter body without any effect
d) none of the mentioned

10. The operation of a machine that utilizes the internal energy of only one TER, thus violating second law is called
a) PMM0
b) PMM1
c) PMM2
d) none of the mentioned

REFRIGERATION AND HEAT PUMP

11. What does a refrigerant do?
a) absorbs the heat leakage into body from surroundings
b) evaporates in the evaporator
c) absorbs latent heat of vaporization form the body which is cooled
d) all of the mentioned

12. Coefficient of performance (COP) is defined as
a) heat leakage/work input
b) work input/heat leakage
c) latent heat of condensation/work input
d) work input/latent heat of condensation

13. What is the relation between COP of an ideal heat pump and an ideal refrigerator?
a) COP of pump=COP of refrigerator – 1
b) COP of pump=COP of refrigerator + 1
c) COP of pump=COP of refrigerator – 2
d) COP of pump=COP of refrigerator + 2

14. Kelvin-Planck's and Clausius' statements are
a) not connected to each other
b) virtually two parallel statements of second law
c) violation of one doesn't violate the other
d) none of the mentioned

15. If one of the Kelvin-Planck's or Clausius' statement is violated, then other is also violated.
a) True
b) false

CARNOT THEOREM

16. Carnot cycle is a reversible cycle.
a) True
b) false

17. The correct sequence of the processes taking place in a Carnot cycle is
a) adiabatic -> adiabatic -> isothermal -> isothermal
b) adiabatic -> isothermal -> adiabatic -> isothermal
c) isothermal -> isothermal -> adiabatic -> adiabatic
d) isothermal -> adiabatic -> isothermal -> adiabatic

18. According to Carnot's theorem, all heat engines operating between a given constant temperature source and sink, none has a higher efficiency than a reversible engine.
a) True
b) false

19. For a reversible refrigerator, Coefficient of Performance is given by
a) $T_L/ (T_H-T_L)$
b) $T_H/ (T_H-T_L)$
c) $T_L/(T_L-T_H)$
d) $T_H/(T_L-T_H)$

20. For a reversible heat pump, COP is given by
a) $T_L/(T_H-T_L)$
b) $T_H/(T_H-T_L)$
c) $T_L/(T_L-T_H)$
d) $T_H/(T_L-T_H)$

CLAUSIUS THEOREM

21. The efficiency of a general cycle will be _____ the efficiency of a reversible cycle.
a) equal to
b) less than
c) equal to or greater than
d) equal to or less than

22. If the cyclic integral of $\delta Q/T$ is zero then the cycle is
a) irreversible but not possible
b) irreversible but possible
c) impossible
d) reversible

23. If the cyclic integral of $\delta Q/T$ is greater than zero then the cycle is
a) irreversible but not possible
b) irreversible but possible
c) impossible
d) reversible

24. A heat engine receives 6 kW from a source at 250°C and rejects heat at 30°C with W. = 6 kW. Does this satisfy the inequality of Clausius?
a) Yes
b) No
c) Cannot be said
d) None of the mentioned

PROPERTIES OF PURE SUBSTANCES

PV DIAGRAM

25. For water, as temperature increases, volume always increases?
a) True
b) false

26. In which of the following state(s) does water exist?
a) Saturated solid state
b) saturated liquid state
c) saturated vapour state
d) all of the mentioned

27. The triple point is a line on the p-V diagram, where all the three phases, solid, liquid and gas exist.
a) True
b) false

28. Which of the following statement(s) is/are true?
a) to the left of saturated solid line is the solid region
b) between saturated solid line and saturated liquid line with respect to solidification there exists the solid-liquid mixture region
c) between two saturated liquid lines is the compressed liquid region
d) all of the mentioned

29. The greater the temperature, the ____ is the vapour pressure of a pure substance.
a) Lower
b) higher
c) none of the mentioned
d) all of the mentioned

30. Which of the following statement(s) is/are true?
a) Saturation temperature is a function of pressure
b) saturation pressure is a function of temperature
c) both of the mentioned
d) none of the mentioned

PURE SUBSTANCE

31. Above the critical point, the isotherms are continuous curves.
a) True
b) false

P-T AND T-S DIAGRAM

32. The slopes of sublimation and vaporization curves for all substances are
a) negative
b) positive
c) zero
d) none of the mentioned

33. The temperature at which a liquid boils is very sensitive to pressure but the temperature at which a solid melts is not such a strong function of pressure.
a) True
b) false

34. The equation which forms the basis of the Mollier diagram is
a) Tds=-dh + vdp
b) Tds=dh + vdp
c) Tds=dh - vdp
d) none of the mentioned

35. Which of the following represents the latent heat of vaporization at a particular pressure.
a) H_f-H_g
b) H_g-H_f
c) H_f +H_g
d) none of the mentioned

36. In the Mollier diagram, the constant pressure lines diverge from one another.
a) True
b) false

DRYNESS FRACTION

37. Quality indicates the
a) mass fraction of liquid in a liquid vapour mixture
b) mass fraction of vapour in a liquid vapour mixture
c) both of the mentioned
d) none of the mentioned

38. Which of the following statements is true?
a) The value of x varies between 0 and 1
b) for saturated water, x=0
c) for saturated vapour, x=1
d) all of the mentioned

39. Voidage is given by
a) specific volume of saturated vapour / specific volume of liquid vapour mixture
b) specific volume of liquid vapour mixture / specific volume of saturated vapour
c) specific volume of saturated liquid / specific volume of liquid vapour mixture
d) specific volume of liquid vapour mixture / specific volume of saturated liquid

40. Which of the following is correct?
a) $v = v_f + (x * v_{fg})$
b) $h = h_f + (x * h_{fg})$
c) $s = s_f + (x * s_{fg})$
d) all of the mentioned

STEAM TABLES

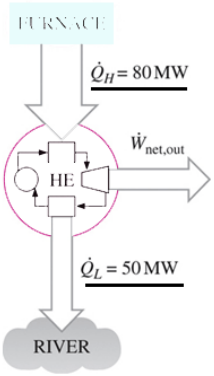
41. The internal energy of saturated water at the triple point is
a) 1
b) 0
c) -1
d) infinity

42. Saturated liquid or the saturated vapour has how many independent variables?
a) One
b) two
c) three
d) none of the mentioned

43. When does a vapour become superheated?
a) when the temperature of vapour is less than the saturation temperature at given pressure
b) when the temperature of vapour is more than the saturation temperature at given pressure
c) when the temperature of vapour is equal to the saturation temperature at given pressure
d) none of the mentioned

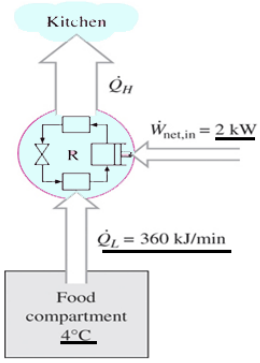
THERMODYNAMICS LAW

44. Heat is transferred to a heat engine from a furnace at a rate of 80 MW. If the rate of waste heat rejection to a nearby river is 50 MW, determine the thermal efficiency for this heat



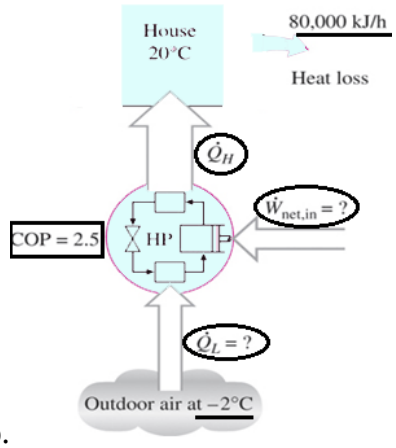
engine.
a) 47.5 %
b) 27.5 %
c) 37.5 %
d) none of the mentioned

45. The food compartment of a refrigerator is maintained at 4 °C by removing heat from it at a rate of 360 kJ/min. If the required power input to the refrigerator is 2 kW, determine the coefficient of performance of the refrigerator.



a) 4
b) 3
c) 2
d) 1

46. A heat pump is used to meet the heating requirements of a house and maintain it at 20 °C. On a day when the outdoor air temperature drops to 2 °C, the house is estimated to lose heat at a rate of 80,000 kJ/h. If the heat pump under these conditions has a COP of 2.5, determine the power consumed by the heat pump.



a) 32000 kJ/h
b) 33000 kJ/h
c) 34000 kJ/h
d) 35000 kJ/h

47. An air-conditioner provides 1 kg/s of air at 15 °C cooled from outside atmospheric air at 35°C. Estimate the amount of power needed to operate the air-conditioner.

a) 1.10 kW
b) 1.20 kW
c) 1.30 kW
d) 1.40 kW

48. In a cryogenic experiment you need to keep a container at -266.84 °C although it gains 100 W due to heat transfer. What is the smallest motor you would need for a heat pump absorbing heat from the container and rejecting heat to the room at 20 °C?
a) 97.85 kW
b) 98.85 kW
c) 99.85 kW
d) 95.85 kW

49. A piston –cylinder device contains 0.06 m^3 of saturated water vapour at 350 kPa pressure. Determine the mass of the vapour inside the cylinder.
a) 0.104 kg
b) 0.124 kg
c) 0.134 kg
d) 0.114 kg

50. A rigid tank contains 10 kg of water at 90 °C. If 8 kg of the water is in the liquid form and the rest is in the vapour form, determine the volume of the tank.
a) 1.73 m^3
b) 2.73 m^3
c) 3.73 m^3
d) 4.73 m^3

51. An 80-litre vessel contains 4 kg of R-134a at a pressure of 160 kPa. Determine the quality.
a) 0.127
b) 0.137
c) 0.147
d) 0.157.

52. Determine the specific volume of R-134a at 1 MPa and 50 °C, using ideal gas equation of state. Take R = 0.0815 kJ/kg K
a) 0.022329 m^3/kg
b) 0.026369 m^3/kg
c) 0.025329 m^3/kg
d) 0.026329 m^3/kg

53. A rigid tank contains 50 kg of saturated liquid water at 90 °C. Determine the volume of the tank.
a) 0.0518 m^3
b) 0.0618 m^3
c) 0.0718 m^3
d) 0.0818 m^3

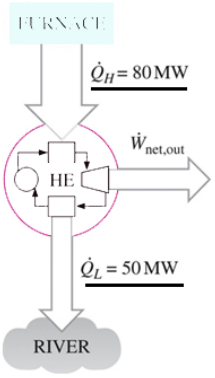
54. A rigid tank contains 10 kg of water at 90 °C. If 8 kg of the water is in the liquid form and the rest is in the vapour form, determine the pressure in the tank.
a) 60.183 kPa
b) 70.183 kPa
c) 80.183 kPa
d) 90.183 kPa

55. An 80-litre vessel contains 4 kg of R-134a at a pressure of 160 kPa. Determine the temperature.
a) -10.60 °C
b) -13.60 °C
c) -15.60 °C
d) -19.60 °C

56. An 80-litre vessel contains 4 kg of R-134a at a pressure of 160 kPa. Determine the volume occupied by the vapour phase.
a) 0.0775 m^3
b) 0.0575 m^3
c) 0.0975 m^3
d) 0.0375 m^3

THERMODYNAMICS LAW

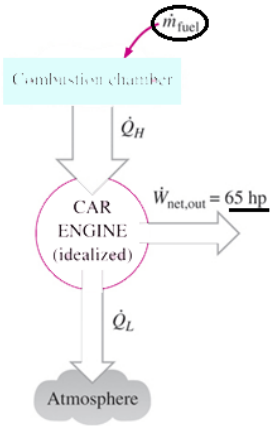
57. Heat is transferred to a heat engine from a furnace at a rate of 80 MW. If the rate of waste heat rejection to a nearby river is 50 MW, determine the net power output for this heat



engine.

a) 30 MW
b) 40 MW
c) 50 MW
d) 60 MW

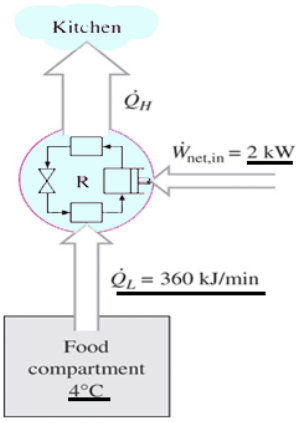
58. A car engine with a power output of 50 kW has a thermal efficiency of 24 percent. Determine the fuel consumption rate of this car if the fuel has a heating value of 44,000 kJ/kg



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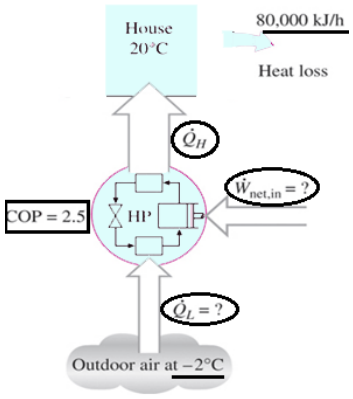
a) 0.00273 kg/s
b) 0.00373 kg/s
c) 0.00473 kg/s
d) 0.00573 kg/s

59. The food compartment of a refrigerator is maintained at 4 °C by removing heat from it at a rate of 360 kJ/min. If the required power input to the refrigerator is 2 kW, determine the rate of heat rejection to the room that houses the refrigerator.



a) 450 kJ/min
b) 460 kJ/min
c) 470 kJ/min
d) 480 kJ/min

60. A heat pump is used to meet the heating requirements of a house and maintain it at 20 °C. On a day when the outdoor air temperature drops to 2 °C, the house is estimated to lose heat at a rate of 80,000 kJ/h. If the heat pump under these conditions has a COP of 2.5, determine the rate at which heat is absorbed from the cold outdoor air.



a) 32000 kJ/h
b) 48000 kJ/h
c) 54000 kJ/h
d) 72000 kJ/h

61. A cyclic machine, as shown below, receives 325 kJ from a 1000 K energy reservoir. It rejects 125 kJ to a 400 K energy reservoir and the cycle produces 200 kJ of work as output. Is this cycle reversible, irreversible, or impossible?
a) Reversible
b) irreversible
c) impossible
d) none of the mentioned

62. A car engine operates with a thermal efficiency of 35%. Assume the air-conditioner has a coefficient of performance of 3 working as a refrigerator cooling the inside using engine shaft work to drive it. How much fuel energy should be spend extra to remove 1 kJ from the inside?
a) 0.752 kJ
b) 0.952 kJ
c) 0.852 kJ
d) none of the mentioned

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